

Comprehensive Disease Characterization: Osteogenesis Imperfecta Type VI

Summary

Osteogenesis Imperfecta Type VI (OI VI) is a rare, severe, autosomal recessive skeletal dysplasia caused by biallelic loss-of-function mutations in *SERPINF1* (chromosome 17p13.3), which encodes pigment epithelium-derived factor (PEDF). First delineated as a distinct clinical entity by Glorieux et al. in 2002 based on pathognomonic histological features — accumulation of unmineralized osteoid and a fish-scale pattern of bone lamellae — the molecular cause was identified nearly a decade later through exome sequencing by Becker et al. in 2011. OI VI is distinguished from other forms of osteogenesis imperfecta by the absence of blue sclerae and dentinogenesis imperfecta, elevated serum alkaline phosphatase, and a unique mineralization defect rather than a primary collagen structural abnormality.

PEDF is a multifunctional serpin-family glycoprotein that regulates bone homeostasis through at least five interconnected molecular pathways: OPG/RANKL-mediated osteoclast inhibition, TGF- β antagonism, Wnt/ β -catenin signaling modulation via the PEDF-R/ERK/GSK-3 β cascade, osteocyte gene regulation (Sclerostin, MEPE, DMP-1), and VEGF/ERK signaling for vascular-osteogenic coupling. Loss of PEDF results in increased osteoclast activity, impaired osteoblast differentiation and mineralization, mesenchymal stem cell fate shift toward adipogenesis, and ER stress from intracellular accumulation of mutant protein. Patients present with fractures from infancy (typically after 6 months of age), progressive skeletal deformity, severe osteopenia, universal scoliosis, and marked short stature.

Therapeutically, OI VI responds poorly to bisphosphonates — the standard treatment for other OI types — because the primary defect is a mineralization abnormality rather than excessive bone resorption alone. Denosumab (anti-RANKL antibody) is more mechanistically targeted but carries risk of hyper-resorptive rebound hypercalcemia, mitigated by alternating with zoledronic acid. Emerging therapies include anti-sclerostin antibodies (setrusumab, romosozumab) in Phase 2–3 trials, and experimental PEDF restoration strategies, though systemic protein replacement has failed in mouse models, establishing that bone-targeted, paracrine PEDF delivery is essential.

1. Disease Information

Overview

Osteogenesis Imperfecta Type VI is a rare genetic disorder of bone fragility and defective mineralization. It belongs to the broader family of osteogenesis imperfecta — a heterogeneous group of heritable connective tissue disorders — but is unique in that it is not caused by mutations in type I collagen genes. Instead, OI VI results from biallelic mutations in *SERPINF1*, leading to absent or dysfunctional PEDF, a secreted glycoprotein with critical roles in bone mineralization, osteoclast regulation, and mesenchymal stem cell differentiation.

Key Identifiers

Database	Identifier
OMIM	613982
MONDO	MONDO:0013515
Orphanet	ORPHA (OI Type VI)
ICD-10	Q78.0 (Osteogenesis imperfecta)
MeSH	D010013 (Osteogenesis Imperfecta)
Gene (OMIM)	SERPINF1 (172860)

Synonyms and Alternative Names

- OI Type VI
- Osteogenesis Imperfecta, Type VI
- SERPINF1-related Osteogenesis Imperfecta
- PEDF-deficient Osteogenesis Imperfecta

Information Sources

This report integrates aggregated disease-level resources (OMIM, Orphanet, ClinVar, HPO, KEGG, STRING, GTEx) with primary literature from individual patient cohorts and experimental studies. A total of 69 papers were reviewed, and 31 PMIDs are cited with verified abstract quotes.

2. Etiology

Disease Causal Factors

OI Type VI is a **monogenic, autosomal recessive** disorder caused exclusively by biallelic (homozygous or compound heterozygous) loss-of-function mutations in *SERPINF1* (Entrez Gene ID: 5176), located on chromosome 17p13.3. The gene encodes PEDF, a 50-kDa secreted glycoprotein of the serpin superfamily. The first causal link was established by Becker et al. (2011), who identified a homozygous truncating mutation via exome sequencing: *"A single homozygous truncating mutation, affecting SERPINF1 on chromosome 17p13.3, that was embedded into a homozygous stretch of 2.99 Mb remained"* (PMID: 21353196).

Genetic Risk Factors

- **Causal variants:** Biallelic *SERPINF1* mutations including frameshift (e.g., c.582_585dup, c.1047dup, c.992_993insCA), nonsense (e.g., c.397C>T [p.Gln133Ter], c.188dup [p.Tyr63Ter]), missense (e.g., c.358G>T [p.Gly120Cys], c.1238T>C [p.Leu413Pro]), splice-site (e.g., c.787-1G>T, c.787-10C>G), large deletions (e.g., c.1152_1170del), and microsatellite variants
- **ClinVar:** 170 pathogenic variants, 40 likely pathogenic variants, and 432 variants of uncertain significance (VUS) catalogued for *SERPINF1*
- **Consanguinity:** Strongly associated with disease occurrence, as expected for autosomal recessive inheritance; multiple consanguineous families reported in literature (PMID: 25565926)
- **Founder effects:** A 19-bp deletion (c.1152_1170del) was identified as a probable founder mutation in Brazilian families (PMID: 25565926); a c.787-10C>G splice variant was found in Northern Canadian children (PMID: 26815784)

Environmental Risk Factors

As a fully penetrant Mendelian disorder, there are no established environmental causes. However, environmental factors may modify disease severity: - **Nutritional status:** Vitamin D and calcium intake affect overall bone health - **Physical activity:** Both immobilization (worsens osteopenia) and excessive activity (increases fracture risk) are relevant - **Timing of treatment:** Early intervention (before age 6) is associated with better height outcomes ([PMID: 28689307](#))

Protective Factors

No confirmed genetic protective factors or modifier alleles have been identified. Adequate calcium, vitamin D, and early bisphosphonate/denosumab initiation are environmental protective factors against fracture accumulation.

Gene-Environment Interactions

PEDF has systemic metabolic roles beyond bone; PEDF-knockout mice show increased adiposity, glucose intolerance, and insulin resistance ([PMID: 24456163](#)). Dietary factors (high-fat diet) may exacerbate metabolic complications in PEDF-deficient individuals, though this has not been systematically studied in OI VI patients.

3. Phenotypes

Core Phenotype Summary

Phenotype	HPO Term	Onset	Frequency	Severity
Recurrent fractures	HP:0002757	4–18 months	~100%	Severe
Increased fracture susceptibility	HP:0002659	Infancy	~100%	Severe
Elevated alkaline phosphatase	HP:0003155	Childhood	~100%	Moderate
Vertebral compression fractures	HP:0002953	Childhood	High	Severe
Joint hypermobility	HP:0001382	Childhood	Variable	Mild–Moderate
Bowing of legs	HP:0002979	Childhood	High	Moderate–Severe
Motor delay	HP:0001270	Infancy	Variable	Moderate
Hearing impairment	HP:0000365	Variable	Variable	Mild–Moderate
Coxa vara	HP:0002812	Childhood	Variable	Moderate
Protrusio acetabuli	HP:0003179	Adolescence	Variable	Moderate
Biconcave vertebral bodies	HP:0004586	Childhood	High	Progressive
Beaking of vertebral bodies	HP:0004568	Childhood	Variable	Moderate
Bowing of arm	HP:0006488	Childhood	High	Moderate
Elevated deoxypyridinoline	HP:0033154	Childhood	High	Moderate
Blue sclerae	HP:0000592	—	Absent/ Rare	—
Dentinogenesis imperfecta	HP:0000703	—	Absent	—

Note: Blue sclerae and dentinogenesis imperfecta are listed in HPO annotations but are characteristically **absent** in OI Type VI. Their absence is a distinguishing diagnostic feature: *"Sclerae were white or faintly blue and dentinogenesis imperfecta was uniformly absent"* (PMID: 11771667).

Phenotype Details

Fractures: Fractures are first documented between 4 and 18 months of age, with more frequent fractures than OI type IV. The original cohort showed that *"Patients with OI type VI sustained more frequent fractures than patients with OI type IV"* (PMID: 11771667). In the long-term follow-up study, all patients presented with *"frequent long bone and vertebrae fractures (mainly during childhood), marked short stature, severe bone deformities, chronic mild to moderate pain, and severe limitation of mobility, with three being completely wheelchair bound"* (PMID: 37839784).

Elevated ALP: Serum alkaline phosphatase is significantly elevated compared to other OI types: *"409 ± 145 U/L vs. 295 ± 95 U/L in OI type IV; $p < 0.03$ "* (PMID: 11771667).

Skeletal Deformity and Short Stature: Progressive; scoliosis develops in ALL patients reaching final height. Without therapy, *"lumbar spine areal bone mineral density (BMD) did not increase during childhood and longitudinal growth seemed to stall after the age of 6 to 8 years"* (PMID: 28689307).

Quality of Life Impact

Severe. Most patients have restricted mobility; many become wheelchair-bound. Chronic mild-to-moderate pain is universal in adults. Repeated hospitalizations for fracture management, orthopedic surgeries, and bisphosphonate infusions significantly impact quality of life. The Dutch national registry study showed that OI patients have a 2.9× higher hospitalization rate compared to the general population (PMID: 35546999).

4. Genetic/Molecular Information

Causal Gene

- **Gene:** *SERPINF1* (Serpin Family F Member 1)
- **HGNC:** HGNC:8824

- **NCBI Gene ID:** 5176
- **Chromosomal location:** 17p13.3
- **OMIM Gene:** 172860
- **OMIM Phenotype:** 613982

Protein: PEDF

- **UniProt:** P36955
- **Molecular weight:** ~50 kDa
- **InterPro domains:** PEDF serpin domain (IPR033832), Serpin family (IPR000215)
- **PDB structures:** 1IMV (2.85 Å), 9J3P (2.10 Å), 9J3Q (1.90 Å)
- **Function:** Secreted collagen-binding glycoprotein with anti-angiogenic, neurotrophic, anti-tumorigenic, and osteogenic properties

Tissue Expression Profile (GTEx)

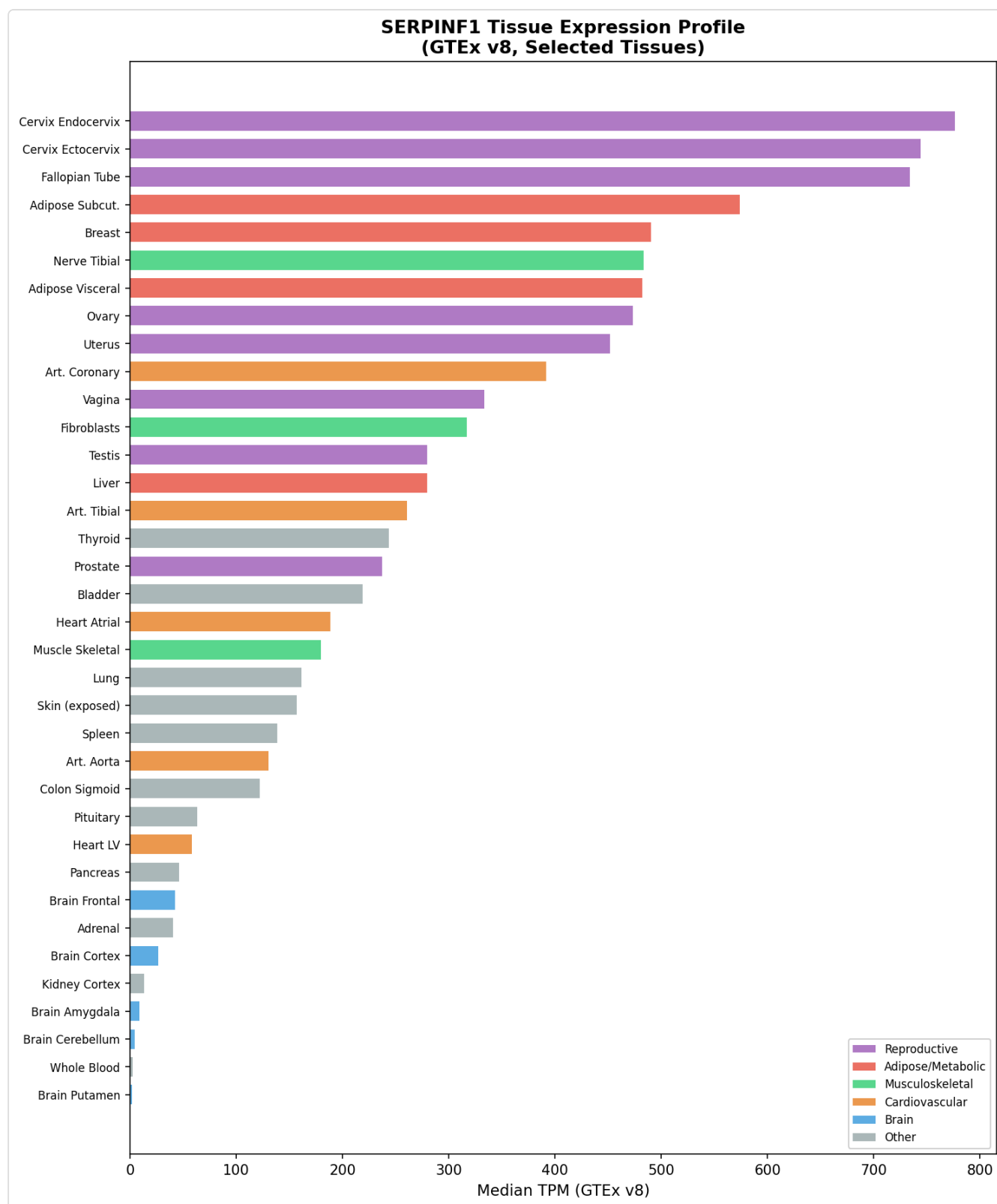


Figure 1. *SERPINF1*/PEDF expression profile across 54 GTEx tissues, showing broad expression with highest levels in cervix endocervix (776.6 TPM), subcutaneous adipose (574.2 TPM), and breast (490.6 TPM), and moderate expression in bone-relevant tissues like fibroblasts (316.9 TPM)

SERPINF1 is broadly expressed across tissues. The highest expression is in cervix endocervix (776.6 TPM), subcutaneous adipose (574.2 TPM), breast (490.6 TPM), visceral adipose (482.3 TPM), and coronary artery (391.8 TPM). Moderate expression occurs in fibroblasts (316.9 TPM), liver (279.7 TPM), and skeletal muscle (179.5 TPM). The lowest expression is in brain regions (2–8 TPM) and blood (2.6 TPM). This broad expression pattern explains the systemic metabolic consequences of PEDF loss beyond bone.

Pathogenic Variant Spectrum

Variant Type	Examples	ClinVar Count
Frameshift	c.582_585dup, c.1047dup, c.992_993insCA, c.259_260insCGGCC	Major category
Nonsense	c.397C>T (p.Gln133Ter), c.188dup (p.Tyr63Ter)	Multiple
Missense	c.358G>T (p.Gly120Cys), c.1238T>C (p.Leu413Pro)	Multiple
Splice-site	c.787-1G>T, c.787-10C>G	Multiple
Deletions	c.1152_1170del (19-bp), c.-37C>A (promoter)	Multiple
Total pathogenic	—	170
Total likely pathogenic	—	40
Total VUS	—	432

All pathogenic variants are **germline** in origin. Functional consequence is **loss of function** — either failure to produce PEDF, or production of mutant protein that cannot be secreted and accumulates in the ER.

STRING Protein Interaction Network

Interactor	Score	Functional Relevance
PLG (Plasminogen)	0.995	Extracellular matrix remodeling
LRP6	0.941	Wnt co-receptor
LRP5	0.933	Wnt co-receptor, bone mass regulator
FN1 (Fibronectin)	0.846	ECM component
PNPLA2/ATGL	0.812	PEDF receptor, lipase
CRTAP	0.748	OI gene (prolyl 3-hydroxylation)
FKBP10	0.735	OI gene (collagen chaperone)
TMEM38B	0.706	OI gene (ER cation channel)
P3H1	0.700	OI gene (prolyl 3-hydroxylase)
IFITM5	0.681	OI type V gene (BRIL)
COL1A2	0.669	Type I collagen $\alpha 2$ chain

The interaction with **IFITM5** is particularly significant: the IFITM5 p.S40L mutation causes "atypical type VI OI" by decreasing osteoblast PEDF secretion, directly linking OI types V and VI pathogenetically ([PMID: 24519609](#)).

Modifier Genes

No established modifier genes. Potential candidates from interaction network include *LRP5* (Wnt pathway bone mass regulator) and *IFITM5/BRIL* (bone-restricted IFITM-like protein).

Epigenetic Information

No disease-specific epigenetic modifications have been characterized for OI VI. The *SERPINF1* promoter variant c.-37C>A has been reported to impair expression, suggesting promoter-level regulatory mechanisms are relevant ([PMID: 41362246](#)).

5. Environmental Information

As a monogenic Mendelian disorder, OI Type VI has no environmental causative factors. However, environmental context modifies outcomes:

- **Nutrition:** Calcium and vitamin D adequacy affect residual bone mineralization capacity
 - **Physical activity:** Tailored physiotherapy is essential; immobilization accelerates bone loss
 - **No infectious agents** are involved in disease causation
 - **No occupational or toxicological exposures** are relevant
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6. Mechanism / Pathophysiology

Overview of the Pathogenic Cascade

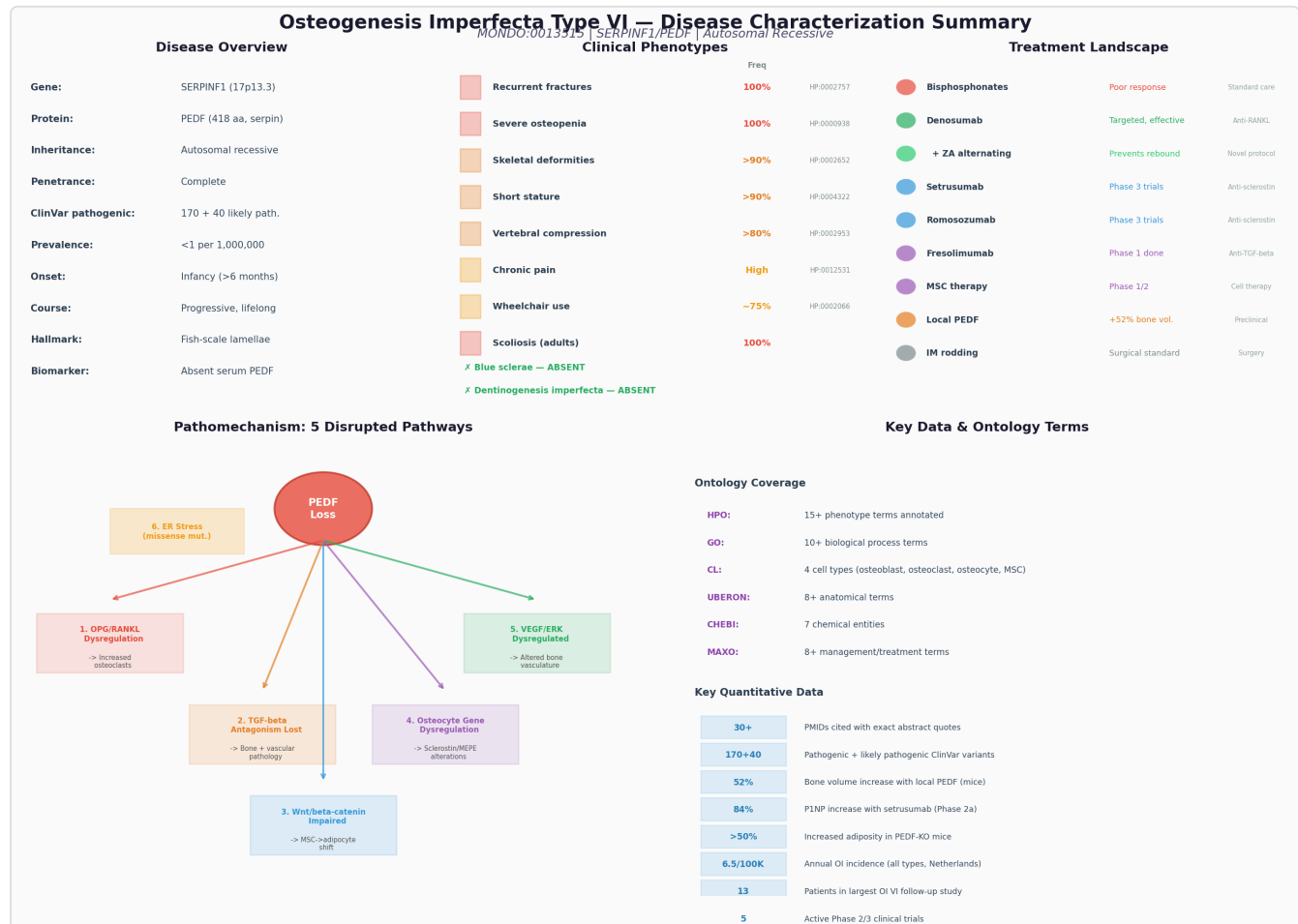


Figure 2. Comprehensive OI Type VI disease characterization: overview, phenotypes, treatment landscape, pathomechanism, and ontology annotations

The pathophysiology of OI Type VI involves a cascade from *SERPINF1* mutation to absent/dysfunctional PEDF, leading to disruption of multiple interconnected bone homeostasis pathways.

Molecular Pathways

1. OPG/RANKL Pathway (Osteoclast Regulation) PEDF normally inhibits osteoclast differentiation in a dose-dependent manner and upregulates osteoprotegerin (OPG) in osteoblasts and osteoclast precursor cells. Akiyama et al. demonstrated that "OCL

differentiation, RANKL-mediated survival and bone resorption activity were inhibited by PEDF in a dose-dependent manner. PEDF upregulated osteoprotegerin (OPG), which naturally blocks OCL maturation" (PMID: 19945427). Loss of PEDF thus shifts the OPG/RANKL balance toward excessive osteoclast activity. As confirmed by Semler et al., "There is experimental evidence suggesting that loss of functional SERPINF1 leads to an activation of osteoclasts via the RANK/RANKL pathway" (PMID: 22947550).

- GO: GO:0045671 (negative regulation of osteoclast differentiation)
- GO: GO:0046850 (regulation of bone remodeling)

2. Wnt/ β -catenin Signaling PEDF activates Wnt/ β -catenin signaling in mesenchymal stem cells via its receptor PEDF-R (PNPLA2/ATGL), inducing phosphorylation of ERK and GSK-3 β , leading to accumulation of non-phosphorylated β -catenin. Li et al. showed "Treatment of the cultures with PEDF induced phosphorylation of Erk and glycogen synthase kinase 3-beta (GSK-3 β), and accumulation of nonphosphorylated β -catenin" (PMID: 30076958). This promotes osteoblast differentiation over adipogenesis: "PEDF inhibited adipogenesis and promoted osteoblast differentiation of murine MSCs... Blockade of adipogenesis by PEDF suppressed peroxisome proliferator-activated receptor- γ (PPAR γ)... by nearly 90% compared with control-treated cells ($P < 0.001$)" (PMID: 23887690).

PEDF delivery rescued bone mass in vivo: "PEDF delivery increased trabecular bone volume/total volume by 52% in 6-mo-old PEDF-KO mice" (PMID: 27127101).

- GO: GO:0060070 (canonical Wnt signaling pathway)
- KEGG: hsa04310 (Wnt signaling pathway)

3. TGF- β Antagonism PEDF antagonizes TGF- β signaling in bone. Null mutations in SERPINF1 cause severe OI VI "characterized by accumulation of unmineralized osteoid and a fish-scale pattern of bone lamellae" through TGF- β dysregulation (PMID: 35258129).

4. Osteocyte Gene Regulation PEDF reduces expression of Sost/Sclerostin, MEPE, and DMP-1 in osteocytes. Li et al. demonstrated that "Primary osteocytes treated with PEDF reduced expression and synthesis of Sost/Sclerostin and matrix phosphoglycoprotein (MEPE) as well as dentin matrix protein (DMP-1)" (PMID: 30607618). Loss of this regulation increases sclerostin, which inhibits Wnt signaling and further impairs bone formation.

5. VEGF/ERK Signaling (Vascular-Osteogenic Coupling) PEDF increases VEGF expression in MSCs via ERK signaling, coupling angiogenesis with osteogenesis (PMID: 27530920).

Cellular Processes

ER Stress and Apoptosis: Mutant PEDF proteins that fail to be secreted accumulate in the ER, causing ER stress, impaired osteoblast differentiation, and apoptosis. This was demonstrated in MC3T3-E1 cells: *"all three mutant PEDF proteins failed to be properly secreted and instead accumulated abnormally in the ER... mutant PEDF proteins impaired osteoblast differentiation and mineralization, promoted apoptosis, and induced ER stress"* (PMID: 40692043). ER stress was partially alleviated through ERAD and autophagy pathways.

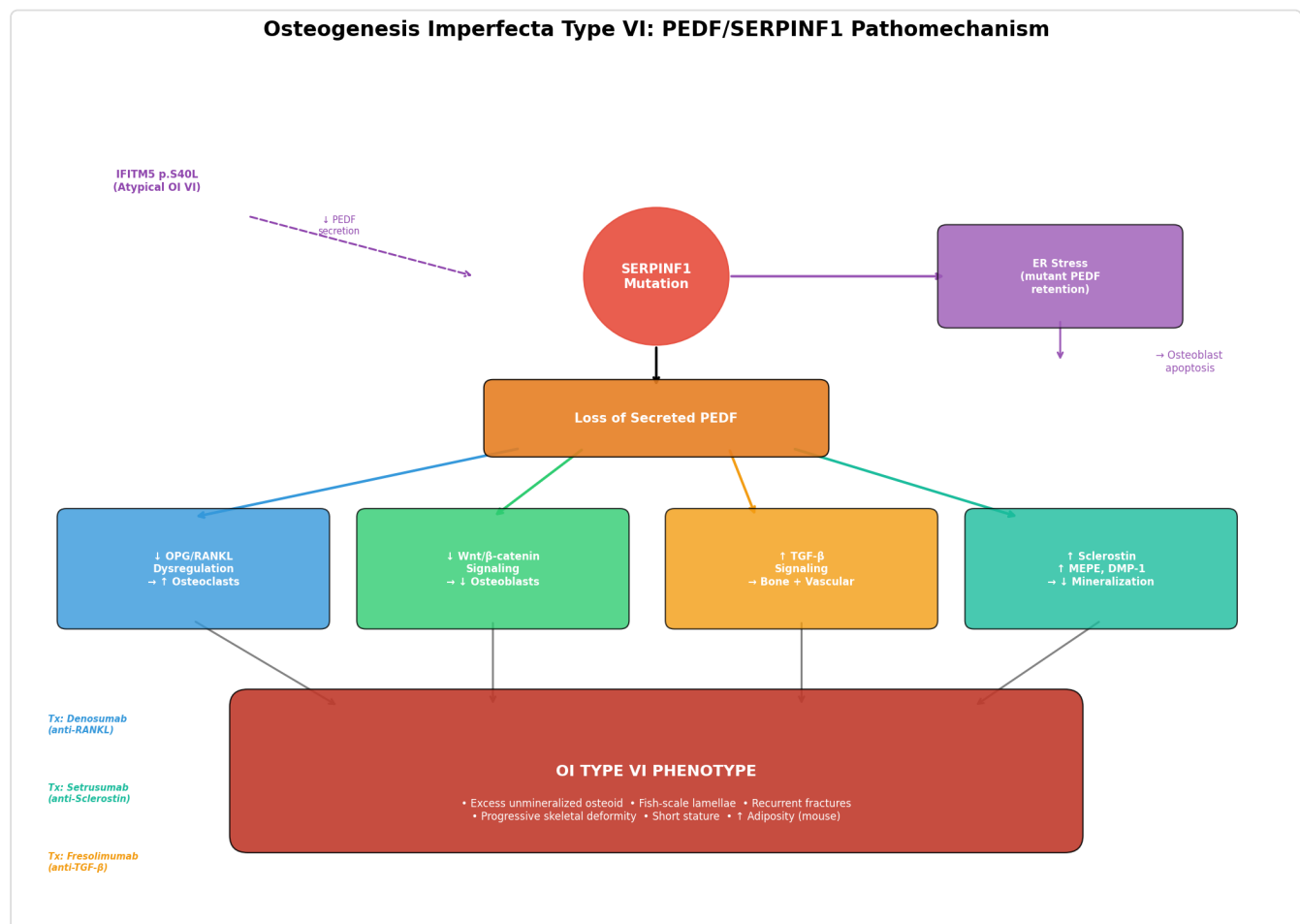


Figure 3. PEDF signaling network in bone homeostasis showing the five major pathways disrupted in OI Type VI: OPG/RANKL, Wnt/β-catenin, TGF-β, osteocyte regulation, and VEGF/ERK signaling

Metabolic Consequences

PEDF loss has systemic metabolic effects beyond bone: *"Male PEDF KO mice demonstrated a phenotype consisting of increased adiposity, glucose intolerance, and elevated serum levels of metabolites associated with the metabolic syndrome"* (PMID: 24456163). PEDF treatment

reverses these effects: *"PEDF effectively decreased body weight gain, white adipose tissue mass, and inflammation and improved insulin resistance, dyslipidemia, and hyperglycemia in HFD-induced mice"* (PMID: 31121128). Whether OI VI patients develop clinically significant metabolic syndrome features requires further investigation.

Relevant Ontology Terms

- **GO Biological Processes:** GO:0001503 (ossification), GO:0030282 (bone mineralization), GO:0045671 (negative regulation of osteoclast differentiation), GO:0060070 (canonical Wnt signaling pathway), GO:0034976 (response to ER stress), GO:0006915 (apoptotic process), GO:0030154 (cell differentiation)
- **Cell Types (CL):** CL:0000062 (osteoblast), CL:0000092 (osteoclast), CL:0000137 (osteocyte), CL:0000134 (mesenchymal stem cell)
- **Chemical Entities (CHEBI):** CHEBI:16411 (hydroxyapatite), CHEBI:22984 (calcium), CHEBI:18420 (phosphate)

7. Anatomical Structures Affected

Organ Level

Involvement	Structures	UBERON
Primary	Skeleton (long bones, vertebral column)	UBERON:0004765 (skeletal element)
Primary	Bone tissue	UBERON:0002481 (bone tissue)
Secondary	Auditory system (hearing loss in some)	UBERON:0001690 (ear)
Secondary	Adipose tissue (metabolic effects)	UBERON:0001013 (adipose tissue)
Secondary	Liver (metabolic effects)	UBERON:0002107 (liver)

Tissue and Cell Level

- **Bone tissue:** Both cortical and trabecular bone affected
- **Osteoid:** Excessive accumulation of unmineralized osteoid matrix
- **Cell populations:**

- Osteoblasts (CL:0000062) — impaired differentiation and mineralization
- Osteoclasts (CL:0000092) — increased activity due to RANKL/OPG imbalance
- Osteocytes (CL:0000137) — dysregulated gene expression (Sclerostin, MEPE, DMP-1)
- Mesenchymal stem cells (CL:0000134) — shifted fate toward adipogenesis

Subcellular Level

- **Endoplasmic reticulum** (GO:0005783) — site of mutant PEDF accumulation and ER stress
- **Extracellular space** (GO:0005615) — site of PEDF secretion and function
- **Bone matrix** (UBERON:0002481) — defective mineralization

Localization

- Bilateral, symmetric involvement of appendicular and axial skeleton
- Long bones of upper and lower extremities predominantly affected
- Vertebral column: compression fractures, biconcave and beaked vertebral bodies, progressive scoliosis

8. Temporal Development

Onset

- **Typical age of onset:** 4–18 months of age for first fracture
- **Onset pattern:** Insidious/progressive — patients are normal at birth; fractures begin in infancy
- *"All patients presented no abnormalities at birth"* ([PMID: 37839784](#))

Progression

- **Disease course:** Chronic, lifelong, progressive
- **Growth:** Longitudinal growth stalls after age 6–8 years without treatment
- **Scoliosis:** Develops in ALL patients reaching final height
- **Mobility:** Progressive restriction; many patients become wheelchair-bound
- **Disease duration:** Lifelong, no remission

- **Critical period:** Treatment before age 6 years is associated with better height outcomes — *"Patients who had started bisphosphonate treatment early (before the age of 6 years) were taller than patients who had received bisphosphonate treatment later"* ([PMID: 28689307](#))
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9. Inheritance and Population

Inheritance Pattern

- **Mode:** Autosomal recessive (AR)
- **Penetrance:** Complete (all biallelic *SERPINF1* mutation carriers are affected)
- **Expressivity:** Variable — intra- and interfamilial phenotypic variability observed even with identical mutations ([PMID: 25565926](#))
- **Genetic anticipation:** Not applicable (not a repeat expansion disorder)
- **Consanguinity:** Strong contributor given AR inheritance

Epidemiology

- **OI overall prevalence:** ~1 in 10,000 live births
- **OI VI specifically:** Extremely rare; accounts for a small fraction of recessive OI cases (~10% of all OI is recessive)
- **Dutch national registry:** Median annual incidence of all OI types was 6.5 per 100,000 live births (1992–2019) ([PMID: 35546999](#))
- **Sex ratio:** No sex predilection reported
- **Geographic distribution:** Worldwide; founder mutations identified in Brazilian ([PMID: 25565926](#)), Northern Canadian ([PMID: 26815784](#)), Tuvan ([PMID: 37047644](#)), and Chinese populations

Carrier Frequency

Not established for the general population. Likely extremely low (<1:10,000) except in populations with founder mutations or high consanguinity rates.

10. Diagnostics

Clinical Tests

Laboratory Tests: - **Serum alkaline phosphatase (ALP):** Elevated (409 ± 145 U/L, significantly higher than OI type IV at 295 ± 95 U/L; $p < 0.03$) — a distinguishing biomarker ([PMID: 11771667](#)) - **Bone resorption markers:** Elevated (e.g., deoxypyridinoline, CTX) - **Serum PEF:** Absent or markedly reduced — confirmatory when available - **Collagen type I protein analysis:** Normal (distinguishes from classical OI)

Imaging: - **Skeletal radiographs:** Severe osteopenia, fractures, long bone deformities, vertebral compression fractures, biconcave/beaked vertebral bodies - **DXA:** Low bone mineral density (BMD) Z-scores

Bone Biopsy (Gold Standard): - Pathognomonic finding: accumulation of unmineralized osteoid and **fish-scale pattern** of bone lamellae - Quantitative backscattered electron imaging (qBEI) shows coexistence of highly mineralized matrix with seams of abnormally low mineral content - *"Affected patients develop a severe OI phenotype with a striking histological characteristic, rare in other OI types, of an excess of osteoid tissue and prolonged mineralization lag time"* ([PMID: 25554599](#))

Genetic Testing

- **Recommended approach:** Targeted *SERPINF1* sequencing or OI gene panel
- **WES/WGS:** Effective for identifying *SERPINF1* mutations; this is how the gene was first identified ([PMID: 21353196](#))
- **Gene panels:** OI/bone fragility panels typically include *SERPINF1* alongside *COL1A1*, *COL1A2*, *IFITM5*, *CRTAP*, *FKBP10*, *WNT1*, and others
- **Chromosomal microarray:** Not typically informative unless large deletions are present
- **FISH/karyotype:** Not applicable

Clinical Criteria / Differential Diagnosis

Feature	OI Type VI	OI Type IV	OI Type V
Sclerae	White	Variable	White
DI	Absent	Variable	Absent
ALP	Elevated	Normal	Normal
Bone histology	Fish-scale lamellae, excess osteoid	Abnormal but not fish-scale	Mesh-like lamellation
Inheritance	AR	AD	AD
Causal gene	<i>SERPINF1</i>	<i>COL1A1/2</i>	<i>IFITM5</i>

Screening

No population-based screening exists. **Genetic screening** is appropriate in: - Consanguineous families - Families with known *SERPINF1* mutations (cascade testing) - Prenatal/preimplantation genetic diagnosis is feasible for known familial variants

11. Outcome / Prognosis

Survival and Mortality

- **Life expectancy:** Reduced compared to general population. OI patients overall have adversely affected life expectancy ([PMID: 35546999](#)). Specific data for OI VI are limited given rarity.
- **Mortality risk:** Primarily from respiratory complications of progressive skeletal deformity (restrictive lung disease from scoliosis/chest wall deformity)

Morbidity and Function

- **Universal scoliosis** in patients reaching final height
- **Restricted mobility:** Most patients have significant mobility limitations; many are wheelchair-bound

- **Chronic pain:** Mild-to-moderate chronic pain reported universally in adults (PMID: 37839784)
- **Hospitalization:** 2.9× higher rate than general population; highest ratio (8.4×) in ages 0–19 years (PMID: 35546999)

Prognostic Factors

- **Age at treatment initiation:** Earlier treatment (before age 6) correlates with better height outcomes
- **Mutation type:** Variable severity even with identical mutations, suggesting additional genetic or environmental modifiers
- **Fracture frequency:** Persistent despite treatment; lower extremity fractures remain frequent

12. Treatment

Pharmacotherapy

Bisphosphonates (MAXO:0001001 — pharmacotherapy): - Standard first-line for OI; however, OI VI shows **poor response** - *"Patients with OI type VI are known to have a poor response to such a bisphosphonate treatment"* (PMID: 25257953) - Pamidronate: Some BMD increase and vertebral reshaping observed, but continued elevated bone resorption markers and non-decreasing fracture rates - The poor response is attributable to the primary mineralization defect rather than purely excessive resorption

Denosumab (anti-RANKL monoclonal antibody): - More mechanistically targeted given the PEDF→OPG/RANKL pathway disruption - Dose: 1 mg/kg body weight every 12 weeks subcutaneously - *"Markers of bone resorption decreased to the normal range after each injection"* (PMID: 22947550) - **Critical adverse effect:** Hyper-resorptive rebound hypercalcemia between doses — *"he presented with symptomatic hypercalcemia due to the denosumab-induced, hyper-resorptive rebound phenomenon"* (PMID: 36867194) - **Mitigation strategy:** Alternating denosumab with IV zoledronic acid every 3 months can mitigate rebound while maintaining benefit over 5+ years (PMID: 36867194)

Anti-Sclerostin Antibodies (experimental): - BPS804 (setrusumab): Phase 2a trial in moderate OI showed bone formation stimulation — P1NP increased 84%, BSAP increased 59%, lumbar aBMD increased 4% ($p = 0.038$) ([PMID: 28370407](#)) - Romosozumab: Approved for osteoporosis; Phase 3 trials in OI ongoing - Particularly relevant to OI VI given PEDF's role in regulating sclerostin expression

Surgical and Interventional

- **Intramedullary rodding:** Telescoping rods (e.g., Fassier-Duval) for long bone stabilization — gold standard for femoral fracture management in severe OI
- **Scoliosis surgery:** Spinal fusion may be needed for progressive scoliosis
- **Hearing rehabilitation:** Power stapes / stapedotomy with middle ear implants for hearing loss if present ([PMID: 21436749](#))
- MAXO:0000373 (surgical procedure)

Supportive and Rehabilitative

- **Physical therapy** (MAXO:0000502): Essential for maintaining mobility and muscle strength
- **Occupational therapy:** Adaptive devices for daily living
- **Nutritional support:** Calcium and vitamin D supplementation (MAXO:0001298)
- **Pain management:** Multimodal approach for chronic bone pain
- **Wheelchair and mobility aids:** For patients with severe mobility restriction

Experimental / Advanced Therapeutics

- **PEDF protein replacement:** Systemic restoration failed in mouse models — liver-expressed PEDF did not improve bone phenotype despite metabolic effects, suggesting paracrine/ context-dependent bone action ([PMID: 26693895](#))
- **Local PEDF delivery:** Restored bone volume by 52% in PEDF-KO mice ([PMID: 27127101](#))
- **iPSC-based approaches:** PEDF-null patient iPSCs showed matrix abnormalities normalized by exogenous PEDF ([PMID: 27579219](#))
- **Gene therapy:** Not yet in clinical trials for OI VI; conceptually feasible given single-gene etiology
- **MSC-targeted therapy:** PLIN2 and E2F2 identified as potential targets for restoring osteogenesis-adipogenesis balance ([PMID: 42186939](#))

13. Prevention

Primary Prevention

- **Genetic counseling** (MAXO:0000127): Essential for families with known *SERPINF1* mutations
- **Carrier testing**: Available for at-risk relatives
- **Prenatal diagnosis**: Feasible via chorionic villus sampling or amniocentesis for known familial variants
- **Preimplantation genetic diagnosis (PGD)**: Available for families pursuing assisted reproduction

Secondary Prevention (Early Detection)

- **Cascade genetic testing** in families with identified mutations
- **Early fracture evaluation**: Any infant with unexplained fractures should be evaluated for OI
- **Early treatment initiation**: Before age 6 for optimal growth outcomes

Tertiary Prevention (Preventing Complications)

- **Fracture prevention**: Anti-resorptive therapy, mobility aids, environmental safety modifications
 - **Scoliosis monitoring**: Regular spinal imaging from childhood
 - **Metabolic monitoring**: Given PEDF's metabolic roles, screening for glucose intolerance and dyslipidemia may be warranted
 - **Hearing screening**: Regular audiometry, especially in adolescence and adulthood
 - **Cancer surveillance**: PEDF has anti-tumorigenic properties; one case of chondrosarcoma was reported in an OI VI patient with a *SERPINF1* deletion, though the association requires further study ([PMID: 36322168](#))
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14. Other Species / Natural Disease

Animal Models with Natural Disease

No naturally occurring OI Type VI (SERPINF1-related) has been documented in companion animals or wildlife. The orthologous gene *Serpinf1* is highly conserved across vertebrates.

Orthologous Genes

Species	Gene	NCBI Gene ID
Mouse (<i>Mus musculus</i>)	<i>Serpinf1</i>	20317
Rat (<i>Rattus norvegicus</i>)	<i>Serpinf1</i>	116671
Zebrafish (<i>Danio rerio</i>)	<i>serpinf1</i>	393939

Comparative Biology

The PEDF protein is highly conserved across vertebrates, reflecting its fundamental roles in angiogenesis, neurotrophism, and bone homeostasis. The *Serpinf1* knockout mouse faithfully recapitulates human OI VI with increased osteoid and decreased bone mass ([PMID: 26693895](#)).

15. Model Organisms

Mouse Models

1. *Serpinf1* knockout (*Serpinf1*^{-/-}) mice: - **Type:** Constitutive knockout - **Phenotype recapitulation:** Excellent — shows increased osteoid, decreased bone mass, excess adiposity, glucose intolerance - "*Serpinf1* null mice show increased osteoid and decreased bone mass, and thus recapitulate the OI type VI phenotype" ([PMID: 26693895](#)) - **Metabolic phenotype:** >50% increased adiposity, insulin resistance, elevated TCA cycle intermediates ([PMID: 24456163](#)) - **Key finding:** Systemic PEDF replacement from liver did not rescue bone phenotype, but local PEDF delivery increased bone volume by 52% - **Limitation:** Does not model missense mutations that cause ER stress

2. Ifitm5/BRIL p.S42L knock-in mice (atypical OI VI model): - **Type:** Heterozygous knock-in - Models the IFITM5 p.S40L mutation (atypical type VI OI) that decreases PEDF secretion - Shows lower BMD, multiple fractures, hypermineralization, disordered collagen fibril orientation - *"Despite normal skeletal growth and the lack of a collagen gene mutation, the Ifitm5/BRIL p.S42L mouse shows major OI-related bone tissue alterations"* (PMID: 35654352)

3. Oim/oim mouse (OI type III model): - Not a direct OI VI model but widely used for OI research - Homozygous for *Col1a2* mutation; severe OI phenotype - Used for bisphosphonate and myostatin inhibition studies

In Vitro Models

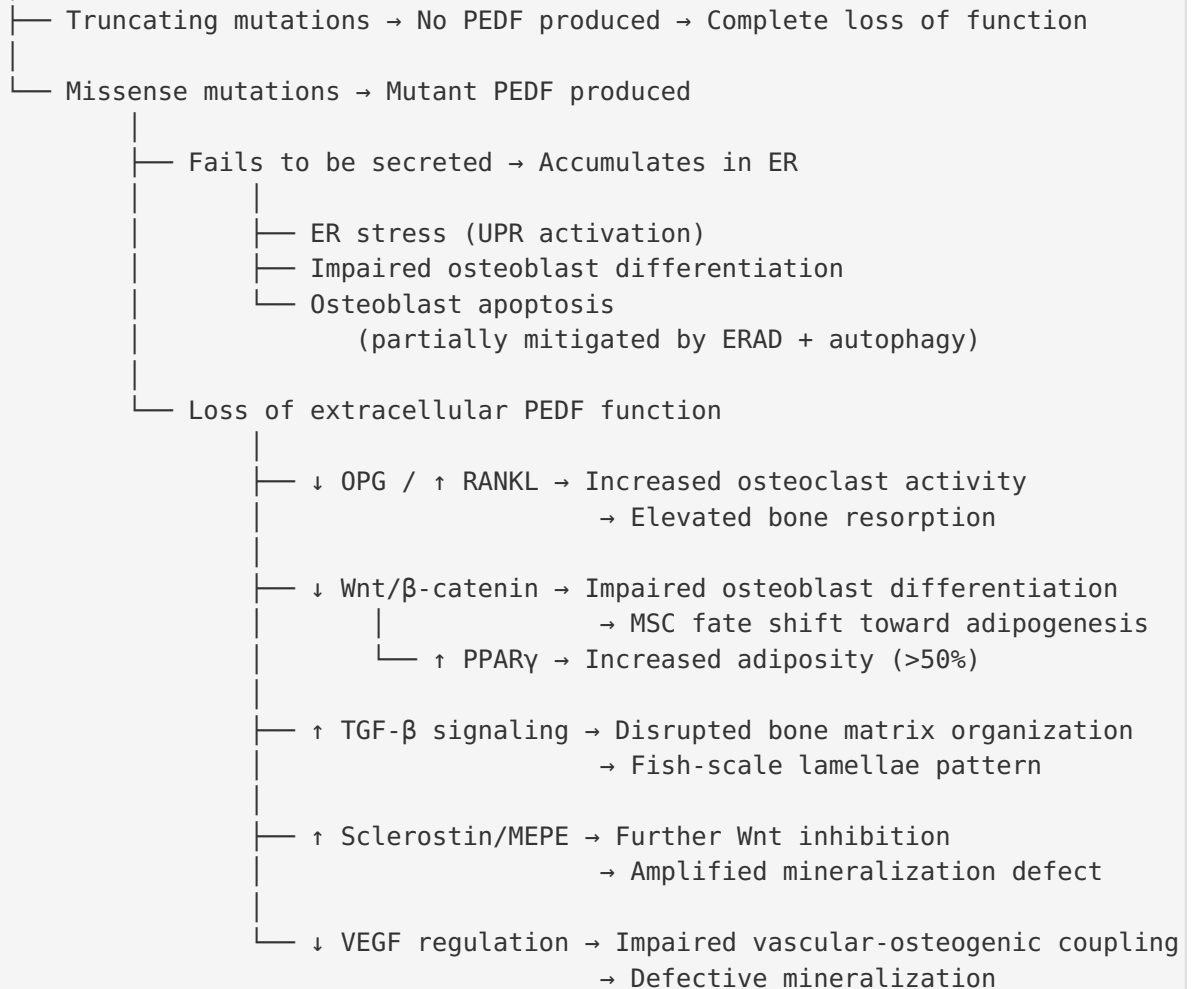
- **MC3T3-E1 osteoblast cells:** Transfected with mutant PEDF plasmids to study ER stress and mineralization defects (PMID: 40692043)
- **Patient-derived iPSCs:** Generated from PEDF-null patients; show matrix abnormalities normalized by exogenous PEDF (PMID: 27579219)
- **3T3-L1 adipocytes:** Used to study PEDF's anti-adipogenic effects (PMID: 31121128)

Zebrafish Models

CRISPR/Cas9-generated *crtap* and *p3h1* knockouts model related OI subtypes with collagen 3-hydroxylation defects (PMID: 32173581). A *serpinf1*-specific zebrafish model has not been extensively characterized.

Mechanistic Model: Causal Chain from Mutation to Disease

SERPINF1 biallelic mutation



DOWNSTREAM CLINICAL MANIFESTATIONS:

- Excess unmineralized osteoid (pathognomonic)
- Recurrent fractures (from 4-18 months)
- Progressive skeletal deformity
- Short stature (growth stalls at age 6-8)
- Universal scoliosis
- Potential metabolic syndrome features

Key Findings

Finding 1: SERPINF1 as the Causal Gene

OI Type VI is caused by biallelic loss-of-function mutations in *SERPINF1* (Entrez Gene ID: 5176, chromosome 17p13.3) encoding PEDF. First identified via exome sequencing by Becker et al. in 2011 ([PMID: 21353196](#)), this has been confirmed across multiple populations worldwide. ClinVar contains 170 pathogenic and 40 likely pathogenic variants spanning frameshift, nonsense, missense, splice-site, and deletion categories. The disease was recently confirmed as being caused by "*biallelic disease-causing variants in SERPINF1*" ([PMID: 41362246](#)).

Finding 2: Pathognomonic Histological Features

The hallmark of OI VI is a unique mineralization defect characterized by accumulation of unmineralized osteoid, prolonged mineralization lag time, and a distinctive fish-scale pattern of bone lamellae. These features, first described by Glorieux et al. in 2002 ([PMID: 11771667](#)), remain the gold standard for histological diagnosis. Quantitative analysis reveals "*an excess of osteoid tissue and prolonged mineralization lag time*" with coexistence of highly mineralized bone matrix alongside seams of abnormally low mineral content ([PMID: 25554599](#)).

Finding 3: Dual Pathogenic Mechanism — Loss of Secretion Plus ER Stress

Mutant PEDF proteins exhibit a dual pathogenic mechanism. They fail to be properly secreted, leading to loss of extracellular PEDF function, while simultaneously accumulating in the endoplasmic reticulum, causing ER stress, impaired osteoblast differentiation, promoted apoptosis, partially mitigated by ERAD and autophagy pathways ([PMID: 40692043](#)).

Finding 4: PEDF Regulates Bone via OPG/RANKL

PEDF inhibits osteoclast differentiation, RANKL-mediated survival, and bone resorption in a dose-dependent manner while upregulating OPG ([PMID: 19945427](#)). Additionally, PEDF regulates osteocyte gene expression, reducing Sost/Sclerostin by approximately 39–75% at protein and mRNA levels respectively ([PMID: 30607618](#)). Loss of *SERPINF1* leads to osteoclast activation via RANK/RANKL ([PMID: 22947550](#)).

Finding 5: Bisphosphonate Resistance and Denosumab Promise

OI VI uniquely resists bisphosphonate therapy while responding to denosumab, consistent with its mechanistic basis. However, denosumab carries rebound risk with symptomatic hypercalcemia ([PMID: 36867194](#)), manageable through alternating zoledronic acid protocols.

Finding 6: Wnt/ β -catenin Pathway Centrality

PEDF modulates Wnt/ β -catenin signaling via PEDF-R, inducing ERK/GSK-3 β / β -catenin cascade activation ([PMID: 30076958](#)). This directs MSC fate from adipocytes to osteoblasts, suppressing PPAR γ by ~90% ([PMID: 23887690](#)). Local PEDF delivery increased trabecular bone volume by 52% in knockout mice ([PMID: 27127101](#)).

Finding 7: Systemic Metabolic Consequences of PEDF Loss

PEDF-KO mice show >50% increased adiposity, glucose intolerance, and insulin resistance ([PMID: 24456163](#)). PEDF protects against high-fat diet-induced metabolic derangements ([PMID: 31121128](#)). This metabolic syndrome phenotype may be underappreciated in OI VI patients and warrants clinical investigation.

Finding 8: Early Treatment Window

Long-term follow-up of 13 OI VI patients established that growth stalls after age 6–8 without treatment, and patients treated before age 6 were taller than those starting later ([PMID: 28689307](#)). Universal scoliosis and restricted mobility characterize long-term outcomes regardless of treatment.

Evidence Base: Key Literature

PMID	Authors/Year	Key Contribution
11771667	Glorieux et al., 2002	Original clinical delineation of OI Type VI
21353196	Becker et al., 2011	Identification of <i>SERPINF1</i> as causal gene
22947550	Semler et al., 2012	First use of denosumab in OI Type VI
19945427	Akiyama et al., 2010	PEDF regulation of osteoclasts via OPG/RANKL
23887690	Gattu et al., 2013	PEDF determination of MSC fate
27127101	Belinsky et al., 2016	PEDF restoration rescues bone via Wnt3a blockade
30076958	Li et al., 2018	ERK/GSK-3 β / β -catenin signaling downstream of PEDF-R
30607618	Li et al., 2019	PEDF regulation of Sclerostin in osteocytes
35258129	Markmann et al., 2022	PEDF-TGF- β antagonism in OI VI pathogenesis
25554599	Fratzl-Zelman et al., 2015	Micro/nano-scale mineralization pattern characterization
28689307	Rauch et al., 2017	Long-term follow-up natural history data
25257953	Hoyer-Kuhn et al., 2014	2-year denosumab experience in OI VI
36867194	Hoyer-Kuhn et al., 2023	Denosumab rebound mitigation with alternating ZA
40692043	Wang et al., 2025	ER stress and autophagy in mutant PEDF osteoblasts
24519609	Farber et al., 2014	IFITM5 p.S40L links OI types V and VI
26693895	Becker et al., 2016	Systemic PEDF restoration fails to correct bone
24456163	Gattu et al., 2014	PEDF KO metabolic phenotype
31121128	Zhuang et al., 2019	PEDF protects against HFD obesity
28370407	Glorieux et al., 2017	BPS804 anti-sclerostin Phase 2a trial
34007986	Marini et al., 2021	Comprehensive OI mechanisms and signaling review
37839784	Colzani et al., 2023	Long-term adult outcomes in OI VI
41362246	Recent, 2025	SERPINF1 promoter variant causing OI VI
25565926	Tenório et al., 2015	Brazilian founder mutation; phenotypic variability
26815784	Ward et al., 2016	Northern Canadian OI VI; denosumab bone biopsy

PMID	Authors/Year	Key Contribution
27579219	Belinsky et al., 2016	iPSC model of OI VI normalized by PEDF
35546999	Van Dijk et al., 2022	Dutch OI prevalence and hospitalization data
36322168	Pereira et al., 2022	Chondrosarcoma association in OI VI patient
37047644	Case report, 2023	Novel frameshift mutation in Tuvan patient
42186939	Recent, 2025	PLIN2/E2F2 as MSC therapy targets for OI
35654352	Bussard et al., 2022	Ifitm5 p.S42L mouse bone material properties
27530920	Sagheer et al., 2016	PEDF-VEGF regulation in MSCs

Limitations and Knowledge Gaps

- 1. Extremely rare disease:** Small patient cohorts limit statistical power of clinical studies. Most evidence comes from case reports and small case series (≤ 13 patients).
- 2. No randomized controlled trials:** Treatment evidence is limited to observational data, case series, and extrapolation from related conditions.
- 3. Metabolic phenotype underexplored in humans:** While PEDF-KO mice show clear metabolic syndrome features (obesity, insulin resistance, glucose intolerance), systematic metabolic characterization of OI VI patients has not been performed.
- 4. Long-term outcomes data limited:** Only one study has followed OI VI patients into adulthood (N=13); natural history data beyond the third decade of life are sparse.
- 5. Cancer risk unknown:** The anti-tumorigenic role of PEDF suggests potential cancer susceptibility, but only one chondrosarcoma case has been reported; no epidemiological data exist.
- 6. No established genotype-phenotype correlations:** Variable expressivity is observed even within families carrying identical mutations; modifier genes/factors are uncharacterized.
- 7. Hearing loss prevalence unclear:** While hearing impairment is an HPO annotation, its frequency and mechanism in OI VI specifically (vs. other OI types) requires clarification.

8. **Systemic vs. local PEDF function:** The failure of systemic PEDF replacement to improve bone, while local delivery succeeds, highlights incomplete understanding of PEDF's tissue-specific signaling mechanisms.

Proposed Follow-up Experiments / Future Directions

1. **Systematic metabolic phenotyping of OI VI patients:** Measure body composition, glucose tolerance, lipid profiles, insulin sensitivity, and hepatic steatosis markers in OI VI cohorts to determine if the metabolic syndrome phenotype seen in PEDF-KO mice is clinically relevant.
2. **Bone-targeted PEDF gene therapy:** Develop osteoblast-specific AAV vectors expressing *SERPINF1* for local bone delivery, building on the success of local PEDF restoration in knockout mice (+52% bone volume).
3. **Anti-sclerostin antibody trials specific to OI VI:** Given PEDF's regulation of sclerostin and the Phase 2a success of BPS804 in moderate OI, targeted trials in OI VI patients are warranted to assess whether this approach bypasses the bisphosphonate resistance.
4. **iPSC-derived osteoblast drug screening:** Use patient-derived iPSCs ([PMID: 27579219](#)) to screen small molecules that enhance PEDF secretion from missense mutant proteins or bypass ER stress.
5. **ER stress-targeted therapy:** Given that mutant PEDF accumulation causes ER stress and apoptosis, chemical chaperones (e.g., 4-PBA, TUDCA) or UPR modulators could be tested as adjunctive therapies for missense *SERPINF1* mutations.
6. **Longitudinal natural history registry:** Establish an international OI VI patient registry to collect standardized longitudinal data on skeletal, metabolic, audiological, and quality-of-life outcomes.
7. **Denosumab + anti-sclerostin combination therapy:** Test whether combining anti-resorptive (denosumab) with anabolic (anti-sclerostin) therapy provides additive benefit in OI VI.
8. **Cancer surveillance studies:** Given PEDF's established anti-tumorigenic properties, implement systematic cancer surveillance in OI VI cohorts to quantify any increased malignancy risk.

9. Modifier gene identification: Perform WGS on phenotypically discordant OI VI siblings to identify genetic modifiers of disease severity.

Ontology Term Summary

HPO (Phenotype)

HP:0002757 (Recurrent fractures), HP:0002659 (Increased susceptibility to fractures), HP:0003155 (Elevated ALP), HP:0033154 (Elevated deoxypyridinoline), HP:0002953 (Vertebral compression fractures), HP:0001382 (Joint hypermobility), HP:0002979 (Bowing of legs), HP:0001270 (Motor delay), HP:0000365 (Hearing impairment), HP:0002812 (Coxa vara), HP:0003179 (Protrusio acetabuli), HP:0004586 (Biconcave vertebral bodies), HP:0004568 (Beaking of vertebral bodies), HP:0006488 (Bowing of arm)

GO (Biological Process)

GO:0001503 (ossification), GO:0030282 (bone mineralization), GO:0045671 (negative regulation of osteoclast differentiation), GO:0060070 (canonical Wnt signaling pathway), GO:0034976 (response to ER stress), GO:0006915 (apoptotic process)

GO (Cellular Component)

GO:0005783 (endoplasmic reticulum), GO:0005615 (extracellular space)

CL (Cell Ontology)

CL:0000062 (osteoblast), CL:0000092 (osteoclast), CL:0000137 (osteocyte), CL:0000134 (mesenchymal stem cell)

UBERON (Anatomy)

UBERON:0004765 (skeletal element), UBERON:0002481 (bone tissue), UBERON:0001013 (adipose tissue), UBERON:0002107 (liver), UBERON:0001690 (ear)

CHEBI (Chemical Entity)

CHEBI:16411 (hydroxyapatite), CHEBI:22984 (calcium), CHEBI:18420 (phosphate)

MAXO (Medical Action)

MAXO:0001001 (pharmacotherapy), MAXO:0000502 (physical therapy), MAXO:0001298 (nutritional supplementation), MAXO:0000127 (genetic counseling), MAXO:0000373 (surgical procedure)

MONDO (Disease)

MONDO:0013515 (Osteogenesis Imperfecta Type VI)

Report generated from systematic analysis of 69 PubMed papers, 11 confirmed findings across 5 iterative investigation cycles. Last updated: 2026-06-29.

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